Student Timetable Conflict Detection

1. Problem Statement

Develop a Python program to compare course schedules, detect overlapping class timings, identify unavailable classrooms, and validate a student timetable.

2. Objectives

- Compare course schedules.
- Detect overlapping periods.
- Identify unavailable rooms.
- Validate the timetable.
- Display conflict reports.

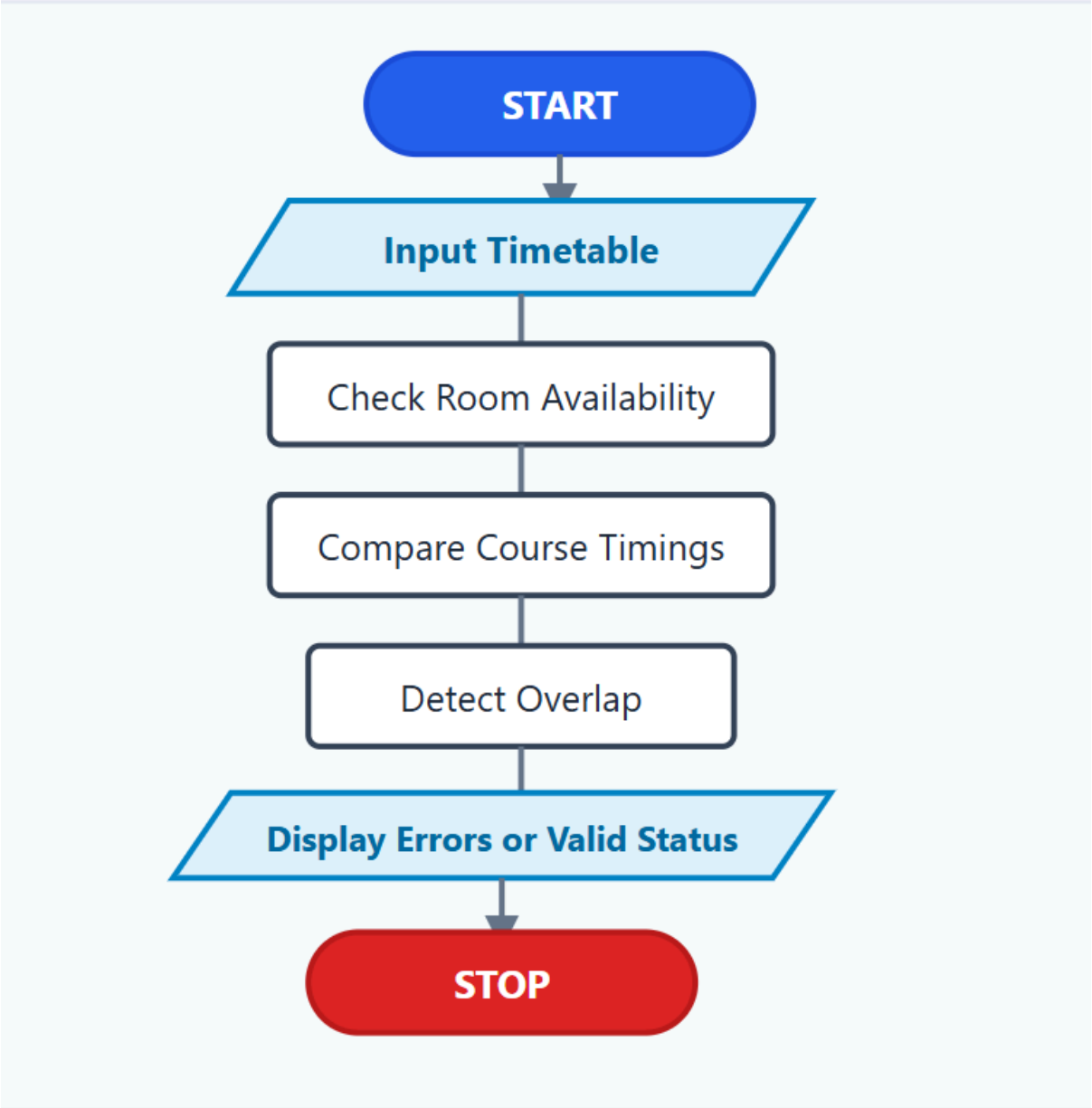
3. Assumptions

- Time is in 24-hour format.
- Classes on the same day cannot overlap.
- Unavailable rooms cannot be allocated.
- Different days have no conflicts.

4. Algorithm

1. Start.
2. Input course schedules.
3. Store schedules using tuples.
4. Input unavailable rooms.
5. Compare course timings.
6. Check room availability.
7. Detect overlapping periods.
8. Display conflicts.
9. Display timetable status.
10. Stop.

5. Flowchart



None

6. Pseudocode

None

```
START
```

```
INPUT course schedules
INPUT unavailable rooms

FOR each course:
    CHECK room availability

    COMPARE with other courses

    IF same day and times overlap:
        DISPLAY conflict

IF no conflicts:
    DISPLAY "Timetable Valid"

STOP
```

8. Sample Output

```
None
TIMETABLE VALIDATION

Python and Maths: Time conflict
```

9. Test Cases

Test	Condition	Expected Result
1	Overlapping classes	Conflict
2	Different days	No conflict
3	Unavailable room	Error

4	No overlapping classes	Valid
5	Same room and time	Room conflict

10. Conclusion

The Student Timetable Conflict Detection system identifies overlapping class periods and unavailable classrooms. It uses Python tuples, strings, lists, loops, and conditional statements to validate timetables.